

An Integrated Framework on Autonomous Valet Parking and Battery Swapping Service Considering Malicious False Data Injection Attack

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Abstract—Autonomous Electric Vehicles (AEVs) have emerged as a transformative technology at the intersection of autonomous driving and electric propulsion. Long-range Autonomous Valet Parking (LAVP) is a promising application for AEVs, catering to diverse parking or drop-off/pick-up demands for varying travel itineraries. However, due to the limited battery capability of AEVs, a reliable energy supplementation strategy for LAVP becomes essential. First, the paper presents battery swapping service as an alternative to traditional plug-in charging, especially for car park in urban area without charging infrastructure. Our work then investigates the coordinated decision-making process for energy supplementation and parking. The decision-making process relies on the battery status of AEVs, and suggests the most appropriate energy supplementation method based on the reserved parking slot's specific type and location. Second, this paper further proposes a detection mechanism to monitor Battery Swapping Station (BSS) status changes and validate data reliability. In case of a malicious attack on BSS, our proposed information recovery method can accurately reflect its actual status with historical arrival information. Third, simulation results indicate desirable performance gains of the proposed scheme, in terms of alleviating congestion on charging service and mitigation on the impact of false data injection attacks.

Index Terms—Autonomous vehicle, Transportation, Information Systems.

NOMENCLATURE

Notations

$x(t)$	System state
$d(t)$	Measurement noise
$e(t)$	Estimation error
$u(t)$	External disturbance
$y(t)$	Battery swapping station measurement value
Λ_{dp}	The set of locations of D/P in network

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$T_{l_{bss}}^{tra}$	Traveling duration from current location to BSS
$\hat{x}(t)$	State estimate value
l_{pl}^{opt}	Optimal parking lot
l_{dp}	D/P point
N_B^0	The number of initial stock battery at BSS
n_B	The number of available batteries
N_C	The queue of batteries being charged
N_D	The queue of depleted batteries removed from AEVs
E^{cur}	Current battery capacity
E^{max}	Maximum capacity of battery
P_0	Rated charging power
ρ^{swap}	The operation time for battery swapping
θ	The total number of charging slots at BSS
EW_{cpu}^{cpu}	Waiting time until a parking slot at CPU
$EWTS$	Expected waiting time for swapping
$T_{cx}^{ae, l_{dp}}$	Traveling duration from current location to D/P point
$T_{cx}^{trip(l_{dp})}$	Outbound (or inbound) trip duration for customer
$T_{cx}^{l_{dp}, wp}$	Customer walking time from D/P point to destination
$T_{l_{bss}}^{tra}$	Traveling duration from current location to BSS

Abbreviations

AEMoD	Autonomous Electric Mobility-on-Demand
AEV	Autonomous Electric Vehicles
AV	Autonomous Vehicles
AVP	Autonomous Valet Parking
LAVP	Long-range Autonomous ValetParking
BSS	Battery Swapping Station
CPR	Car park in Remote Area
CPU	Car park in Urban Area
EV	Electric Vehicle
FDI	False Data Injection Attack
GC	Global Controller
ICT	Information Communication Technology
D/P	Drop-off/Pick-up

I. INTRODUCTION

OWING to rapid advancements in the automotive sector, along with hardware equipment and artificial intelligence (AI), Autonomous Vehicles (AVs) have emerged as a promising solution for fully automated and technologically advanced transportation [1]. Considering the existing drive-by-wire design and in-vehicle system, Electric Vehicles (EVs) have more advantages on autonomous technology implementation [2], [3]. Therefore, the application of Autonomous EVs (AEVs) is progressively supplanting traditional internal combustion engine based AVs. With the increasing popularity

of AEVs across various sectors, there is a growing focus on Autonomous Valet Parking (AVP), enabling car to navigate and park without human intervention.

Most existing studies on AVP have concentrated on car parking operation managed at the entrance [4], [5]. Through laser radar, cameras, and sensors, these vehicles can autonomously navigate to available parking slots, a service typically classified as the Short-range AVP (SAVP). In contrast, recent research has focused on the Long-range AVP (LAVP) [6], [7], extending a broader application scope. Different from SAVP is solely centered on car parking, LAVP prioritizes overall journey experience. In this scenario, customers are dropped off or picked up based on their location preferences, rather than enduring the frustration of waiting at the parking entrance. While LAVP offers appealing characteristics for transportation services, it presents significant thorough long trip, particularly in terms of parking logistics and energy supplementation collaborative decision-making.

Due to the limited battery capacity of AEVs, the efficiency of energy supplementation plays a critical role in influencing public acceptance of AEVs [8]. In response to this challenge, the technology of battery swapping service has been developed. The whole replenishment process only takes 3-5 minutes at Battery Swapping Station (BSS) [9], [10], and can be fully automatically operated through robotics without human intervention. The autonomous functionality of BSS integrates effectively with AEV services, enhancing the progress of Autonomous Electric Mobility-on-Demand (AEMoD) system. The service process of LAVP and AEMoD is similar, but the slight difference is AEMoD system primarily emphasizes timely response to users' service, while the LAVP service intends to minimize charging and parking congestion [11]. Therefore, the existing architectural framework of AEMoD system still offers valuable insights for the design of LAVP system, particularly driven by battery swapping service.

To be specific, inspired by the architectural framework in [11], the BSS and LAVP service are typically operated by two independent systems. Besides, the secure LAVP service protocol has been developed based on ICT in [7], ensuring identity legitimacy and communication security. As for battery swapping service, the importance of information security management and data protection is emphasized in [10]. However, so far, no essential data protection and communication secure methods have been applied in battery swapping service integrated with LAVP. Therefore, it is significant to explore effective coordination strategies between energy supplementation and parking services, taking into account the vulnerabilities in BSS communication and local status data attack.

Motivated by the above issues, our technical contributions can be summarized as follows:

1) *Collaborative decision-making about hybrid energy supplement method and LAVP*: The mechanical parking garage has the remarkable advantages of a small occupying area and a large number of parking spaces, which can provide a new way to alleviate the urban parking issue [12]. Therefore, as depicted in Fig. 1, the Car park in Urban Area (CPU) is modeled as a mechanical parking garage. Due to the unique design structure and limited space, a mechanical parking garage

without charging facilities is prevalent [13]. While, previous works on LAVP management assume that parking lots are all located in suburban area [6], equipped with charging piles and adequate space. Thus, it inspires us to explore car park both in suburban and urban area, so as to jointly determine the optimal parking slot for LAVP process. Besides, considering various charging demands in an urban scenario with AEVs, the battery swapping service emerges as a significant energy supplementation method, alongside traditional plug-in charging. The availability of refueling at BSSs also influences subsequent parking decisions for AEVs, which plays a key role in determining their selection of Car Park in Remote area (CPR) as parking destinations. Therefore, this paper further investigates the collaborative decision-making process concerning hybrid energy supplementation methods and LAVP.

2) *Investigating the impact of malicious attack on BSS status estimation*: Although the reliability of LAVP system integrated with charging service has been guaranteed in [6]–[8], BSSs are often operated by an independent system [11] and involved data vulnerability issues [10]. This may lead to attackers manipulate information through False Data Injection (FDI) attacks on vulnerable channels. Considering effective collaborative decision-making in energy supplementation and parking, it is thus necessary to detect the tampering status information of BSSs and restore the received local status. The proposed detection mechanism mainly consists of boundaries detection, status change detection, and observation detection. Firstly, the boundaries detection method considers the limitation of BSS configuration (such as the total number of available batteries, the total number of charging slots). If the length of queue information exceeds the maximum boundary value, it can be concluded that the data is tampered and BSS is attacked. Secondly, the status change detection method utilizes temporal correlation among status information to assert reliability of adjacent time instant data. Thirdly, the observation detection method defines the AEV that arrives earliest after the service requirement time as “observer”, the expected stay duration of “observer” calculated by sensing data is compared with the actual value, so as to assess the reliability of sensing data. Through above detection steps, if an FDI attack is confirmed at BSS, the information recovery mechanism is executed based on historical arrival data. To be specific, since all batteries at BSS are available at the beginning of system time instant, the battery being charged is from arrived AEVs that required battery swapping service. Therefore, the BSS local status can be derived according to energy update of removed batteries.

Compared with the exiting real-world battery swapping application, like NIO [14] and Gogoro [15], the superiority of proposed framework can be summarized as follows:

- The proposed framework considers the coordination optimization about LAVP service and AEV recharging, so as to jointly optimize the requirement of AEV energy supplement and user drop-off/pick-up.
- The proposed framework provides an FDI attack detection mechanism, so as to discover the BSS with local status information tampering. Then, a BSS local status information recovery mechanism is developed, and thus

the attacked BSS service information can be predicated.

This paper is organized as follows. Related work is reviewed in Section II. Section III presents the system model analysis. Section IV investigates the collaborative decision-making system design. Simulation results are demonstrated in Section V. Section VI concludes the paper.

II. RELATED WORK

A. AVP Service Management

The development of AI, ICT, as well as hardware infrastructure have promoted the progress of AVP. Typically, the AVP service management can be classified into SAVP and LAVP modes. In the realm of current practical applications, the majority of existing research primarily focuses on SAVP, employing techniques such as path planning, computer vision, and perception fusion. Utilizing four crucial control modules, the literature [16] achieved reverse parking accurately without human manual intervention. Different from above single parking lot, a collaborative mapping and autonomous parking system is developed in [17] for multi-story parking garages. While SAVP predominantly concentrates on detailed parking process, the LAVP mode focus on the whole service process management and is becoming a promising research orientation. The concept of LAVP was initially introduced in [18], proposing a foundational service framework to mitigate traffic congestion. Furthermore, the study in [19] developed a reservation-enhanced service scheme that anticipates future parking loads, facilitating the selection of optimal car parks to alleviate parking challenges.

B. Energy Supplement Decision-making Optimization

In the ongoing discourse on AVP service management, a notable gap in most existing literature is the lack of attention to energy supplementation requirements. This oversight is particularly critical in LAVP service management for AEVs, where longer trip durations and limited battery capacities pose significant challenges. Regarding energy supplement for AEVs, the plug-in charging and battery swapping service are two primary methods [9]. In recent years, considering the mobility of vehicles, an increasing number of studies have explored energy supplementation about where to charge [6], [20]. As for the optimization problem about online BSS assignment, the work in [20] utilized a bipartite matching method to minimize BSS service congestion. With the help of reservation information, literature [6] developed an integrated framework on AEV charging and parking management scheme, predicating future service states at CPRs. Eventually, the load balance about AEV charging and parking can be achieved across network.

C. BSS Status Estimation Techniques

BSS status estimation techniques can be mainly classified into optimization theory, data-driven technique and heuristic method. Literature [21] modeled EVs as an open queue, and the battery cycle was in a closed queue. Based on queuing theory analysis, a mixed queuing network model is proposed to depict BSS status. This kind of theoretical optimization method relies on the accurate modeling of BSS service and EV arrival process hypothesis. Considering this disadvantage,

the data-driven approach is applied to describe the status of BSS. A long short-term memory neural network is utilized to predicate EV's service demand and further estimate BSS service status [22]. However, it should be noticed that AI methods work successfully on the condition that plenty of historical data is available. Compared with the above model-driven and data-driven techniques, the service framework based on heuristic method, applied in this work, is believed to efficiently handle realistic and real-time BSS decision optimization problems [23].

TABLE I
THE COMPARISON OF DEFINITE METRICS FOR THRESHOLD-BASED
ATTACK DETECTION METHODS

Literature	Attack Types	Service Scenario	Metrics
Our Scheme	FDI Attack	Battery Swapping Service	Service waiting time; Total service number
[24]	DDoS Attack	Online network service	Entropy value; False negative rates
[25]	TCP SYN Flood	Software defined networking	Normalized entropy; Average attack detection time; Average CPU utilization
[26]	DDoS Attack	Client-server environment	Detection rate; False positive Rate; System precision; Accuracy; Detection delay
[27]	FDI Attack	Power grid	Residuals value

D. Threshold-based Attack Detection Methods

Threshold-based attack detection methods have been widely explored in prior literature [24]–[27], especially for Distributed Denial of Service (DDoS) attacks. For a set of legitimate traffic or attack traffic random flows $X = \{x_1, \dots, x_i, \dots, x_N\}$, the definition of entropy is presented in [28] with $H(X) = -\sum_{i=1}^N p(x_i) \log p(x_i)$. Zhou et al. [24] utilized entropy rate measurement to detect DDoS attacks, and false negative rates defined as $\frac{\text{FalseNegatives}}{\text{FalseNegatives} + \text{TruePositives}}$ to identify performance. Kumar et al. [25] developed detection and mitigation solutions for SYN flood attacks in software-defined networking. Above works are based on a static threshold detection, and therefore cannot adapt to changes in legitimate traffic. Considering the disadvantage, Tsobdjou et al. [26] proposed a dynamic threshold algorithm based on Chebyshev's theorem to detect flooding attacks. The classical statistical measure used to assess the performance of a binary classification test, such as accuracy, precision, etc. Apart from above DDoS attack, the threshold-based attack detection method has also been applied in power grid to handle FDI attacks [27]. A residual value is defined as the difference between observed and predicted value. In Table I, including definite metrics, we compare these literature.

E. Promotion of ICT on AEV service management

The ICT serves as the foundation of service management, playing a crucial role in enhancing the widespread adoption and popularity of AEVs. Literature [29] proposed a real-time AEMoD relocation model based on ICT, and thus shared connected autonomous vehicles can access optimal from control center. With the help of ICT, a communication link that supports anticipated AEVs mobility information (i.e. service reservation) and parking slot status monitoring is established in [30], so as to balance parking demand via price adjustment. To some extent, it is challenging to coordinate large-scale service scheduling for AEVs. Fortunately, the centralized management

approach enables the mobility edge devices (AEV side) simple, and more sophisticated optimization is operated at GC side via information aggregation.

F. Our Motivation

With the progressive development of transportation system, the exploration of LAVP service management is emerging as a promising research area. Given the limited battery capacity of AEVs, although there are a few work paid attention to integrate charging management with AVP, they often overlook the practical city scenario where car parks are deployed across both remote and urban areas. Since BSSs are usually located in urban area [31], compared with remote area parking and charging, AEVs can be immediately refilled at BSS and then conveniently parked in urban area, so as to ensure a timely pick-up response. Therefore, it thus inspires us to investigate the collaborative optimization of parking and energy supplementation driven by battery swapping service. Moreover, to the best of our knowledge, no previous research has investigated the impact of malicious attack on BSS local status information. Whereas, in LAVP service process, it is significant to address the adverse impact of attack for optimal parking and energy supplement decision-making.

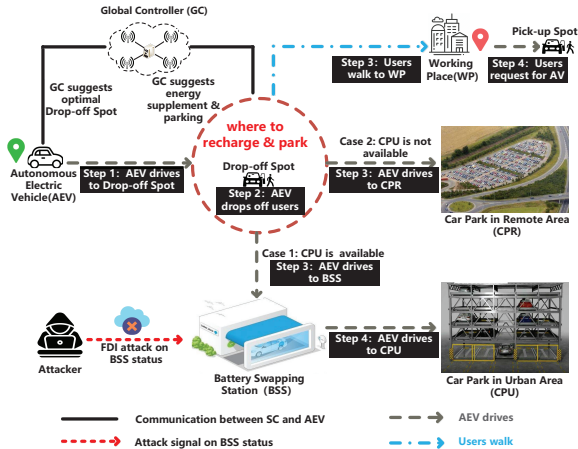


Fig. 1. Big picture of coordinated energy supplement and parking management for AEVs

III. SYSTEM MODEL

A. Overview

Fig. 1 depicts AEV energy supplement and parking coordinated decision-making. The AEV, upon receiving a trip demand from the user, transmits a request to GC determining the optimal drop-off location. Subsequently, the AEV navigates to the recommended spot, where the user is dropped off and walks to Working Place (WP). Once completing drop-off operation, if the AEV's battery level is below the State of Charge (SOC) threshold, it requests energy supplement guidance from the GC. Given that CPU lack charging facilities and BSSs in urban area provide immediate refilling service, the battery swapping service emerges as an essential energy supplement method for AEVs parked in these areas. The comparison between plug-in charging and battery swapping is depicted in Table II. Consequently, the GC preferentially recommends battery swapping service if a CPU is available, and then the AEV drivers to CPU for parking.

TABLE II
THE COMPARISON BETWEEN CHARGING STATIONS AND BATTERY SWAPPING TECHNIQUES

	Charging	Battery Swapping
Duration	4-8 hours	3-5 minutes (*)
Parking	Must park to charge	No limit (*)
User Experience	Poor	Excellent(*)
Battery Safety	No maintenance; Low safety	Unified management and maintenance; High safety (*)
Power Grid Pressure	Disorderly charging behaviour puts huge pressure	Less pressure(*)

This planning is advantageous as it ensures AEVs are strategically parked in urban areas, facilitating timely response to user pick-up requests and reducing waiting times. Compared with EVs that private users own, the LAVP service mode is more like an autonomous commuter bus operated by a third-party company [30]. The main difference between a commuter bus and private cars is that a commuter bus has to pick up/drop off users at designated commuter bus pick-up points and then park at a specific parking area. Compared with commuter buses, private cars are more flexible and can drive directly from the home garage to workplace. The insightful literature [32] investigated consumers' willingness to pay for battery swapping services, and we observed that the cost of battery swapping service is only slightly higher than charging according to NIO company service policy. AEVs are thus preferentially served at BSS and parked at CPU, so as to reduce refilling time and achieve timely responses to customers' pick-up requests. Eventually, the operation company can serve more users, thus improving users' satisfaction, service revenue and acceptance of LAVP via battery swapping service. Whereas, if CPU is not available, AEV drives to a CPR for parking and charging. The following network entities, involved in AEV energy supplement and parking decision-making are introduced:

Autonomous Electric Vehicle (AEV): For each AEV, once the current SOC is below the set threshold, it needs to replenish its battery due to limited capacity. Considering the service experience of customers, the energy supplement process usually happens after AEV dropping off customer at target Drop-off/Pick-up (D/P) point.

Car Park in Remote Area (CPR): CPRs are deployed in the suburban region of a metropolis, thus they are with adequate space to provide enough parking slot for vehicles. Meanwhile, CPRs are equipped with a certain number of charging piles, recharging AEVs while parking. The significant difference between CPR and CPU is that the distance from a CPU to the nearest D/P point is much less than the distance from CPR to its nearest D/P point. Additionally, when AEVs are charged at CPR, the recharging process lasts tens of minutes, during which the AEVs cannot depart.

Car Park in Urban Area (CPU): Different from CPR deployed in suburbs, CPUs are usually located in urban areas, facilitating timely pick-up responses due to desirable locations. Considering the remarkable advantages of a small occupying area and a relatively large number of parking spaces, the mechanical parking garage is widely applied at CPU to alleviate the urban parking issue. As depicted in Fig. 1, due to the lifting

and transmission device of the mechanical parking garage, it cannot provide charging services for AEVs. Therefore, the battery swapping service emerges to solve these dilemmas.

Battery Swapping Station (BSS): Each BSS is equipped with a certain number of stock batteries. After AEVs completing battery swapping operation, depleted batteries will be fully charged in parallel at BSS to meet future demand. Considering the safety for charging batteries at BSS and parking vehicles in CPU, BSSs are positioned at a safe distance from CPU, despite also being located in urban centers.

Global Controller (GC): It is a centralized entity that globally manages all parking and energy supplement demands over whole network.

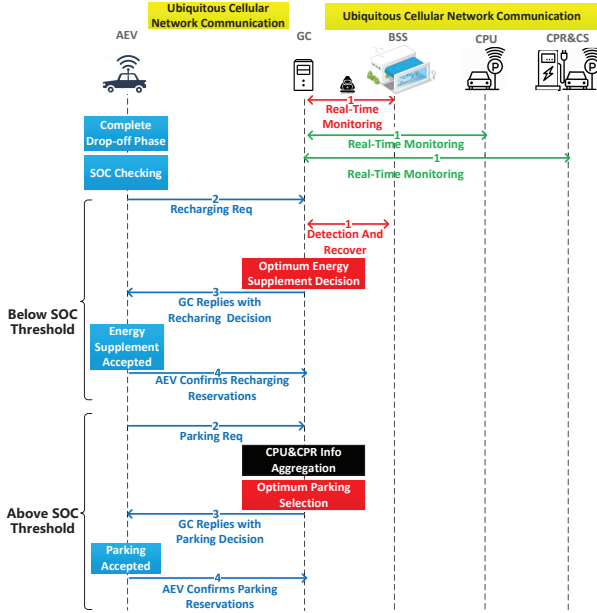


Fig. 2. Time sequences of energy supplement and parking decision-making

B. Protocol for Energy Supplement and Parking

The detailed communication process after AEV completing drop-off phase is depicted in Fig. 2.

Step 1 (Real-time monitoring): The GC continuously monitors charging or parking status of all CPRs and CPUs. The GC also observes the local status of BSS, such as the availability of batteries, queues for battery charging, and queues for depleted batteries.

Upon completing the drop-off phase, the AEV's SOC is assessed, leading to two potential scenarios:

(1) Case 1: Below SOC Threshold

Step 2 (Reporting recharging request): Once an AEV is below the SOC threshold after drop-off phase, it will send the energy supplement request to GC.

Step 3 (Optimum energy supplement decision-making): The GC, utilizing aggregated status information, determines the optimal energy supplementation location and replies this decision to the AEV. Additionally, the GC verifies and, if necessary, corrects the local status of BSS in case of suspected attacks.

For optimum energy supplement decision-making, it means if CPU is available, the optimal BSS is recommended to AEV with minimum total battery swapping service time.

Otherwise, the optimal CPR integrated charging facilities is recommended to AEV with minimum total charging time. As for battery swapping service, it can be described by the following equation:

$$\arg \min_{l_{bss} \in \Lambda_{bss}} (T_{l_{bss}}^{tra} + EWT S_{l_{bss}}) \quad (1)$$

where Λ_{bss} is the set of all BSSs, $T_{l_{bss}}^{tra}$ is the traveling time from current location to BSS, and $EWT S_{l_{bss}}$ is the expected waiting time for swapping. $EWT S_{l_{bss}}$ is defined as:

$$EWT S_{l_{bss}} = \begin{cases} 0, & n_B > 0 \\ ATS_0 - t^{arr}, & \text{otherwise} \end{cases} \quad (2)$$

where n_B is the number of available batteries at BSS, and ATS_0 is the earliest battery charging finishing time.

Step 4 (Confirming optimal recharging decision-making): The AEV will accept the recommendation from GC by sending back its confirmation reply.

(2) Case 2: Above SOC Threshold

Step 2 (Reporting parking request): If the SOC value of AEV is above threshold, the AEV will send parking request (rather than recharging) to GC.

Step 3 (Optimal parking lot selection): In this scenario, the GC collects parking status information from CPRs and CPUs across the network, so as to recommend the most suitable parking lot for the AEV. For optimal parking lot selection, AEV prefers to park at CPU, so as to quickly response users' pick-up requests. The detailed implementation procedure will be elaborated in Section IV-A via Algorithm 1.

Step 4 (Confirming optimal parking decision-making): Similar to the recharging process, the AEV sends a parking reservation confirmation to the GC.

C. The Involved System Communication Protocols and Vulnerability Analysis

The Open Clearing House Protocol (OCHP), the Open Inter Charge Protocol (OICP), the eMobility Inter-operation Protocol (eMIP), and the Open Charge Point Interface (OCPI) are four primary protocols for EV roaming [33], enabling EV accessing GC or any eMobility service provider's charging network. Led by the Open Charge Alliance (OCA), the Open Charge Point Protocol (OCPP) is an open communication protocol, utilized to manage information flow between Electric Vehicle Supplement Equipment (EVSE) and GC, as well as communications between networks themselves. The OCPP coordinates information and power flow between BSSs, global controller, AEVs and power grid. However, due to the clear text characteristic of OCPP, it faces several security threats, such as data breaches and data tampering [34]. Especially, since most BSSs are deployed without consideration of appropriate physical security, their local status information can be accessed and manipulated by attackers [35]. This enables attackers to easily manipulate BSS local status data.

D. The Feasibility of FDI Attack on Battery Swapping Service

1) Vulnerability analysis: With the gradual integration between the Internet of Things (IoTs) and cloud computing, Battery Energy Storage Systems (BESSs) at BSSs have become Cyber Physical Systems based on ICT. This causes severe threats of malicious network attacks on BESSs, among which FDI attackers can tamper with the measured data of batteries collected by sensors, thus affecting the status monitoring

accuracy of BESS [36]. Finally, the actual battery storage and charging status at BSSs cannot be obtained accurately.

2) *Attack pathways*: BESSs consist of three parts: storage batteries, Cloud Battery Management Systems (CBMSs), and bidirectional power conversion systems. When the battery sensor collects battery data, the attacker can inject false data into the control unit. Then, when CBMSs upload the data collected by the battery sensor to cloud storage through the IoTs, the attacker can tamper with the communication data. Based on the structure of BESS system, the attacker can alternately inject small-scale false data through the BMS sensor or communication channel, causing a large deviation in BSS service status, achieving attacking.

The graphic illustration of the typical FDI attack procedure is shown in Fig 3. Besides, the FDI attacker injects tampered data to deceive the global controller, further misleading a decision-making process about optimal BSS-selection. As for real-work attack examples, literature [37] depicted the FDI attack on SOC value of battery packs to achieve battery overcharge or over-discharge, causing permanent damage or even explosion. In 2019, the attack on BESSs caused several battery packs in Korean on fire [38]. The FDI attacker targeted the Ukrainian power grid's control systems, affecting 225,000 users' power demand [39].

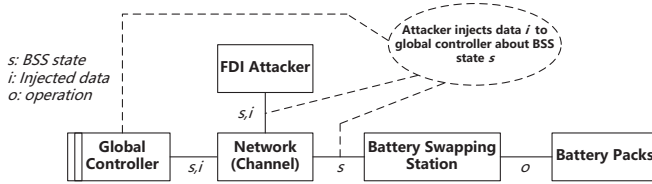


Fig. 3. FDI attack on BSS status information

3) *Cybersecurity measures*: The literature [40] theoretically analyzed the impact of FDI attackers on SOC estimation in BESSs, and proposed a static detection model via the temporal correlation among measured values. From the perspective of attackers, the literature [41] constructed attacks through injecting false data to a subset of the sensors. Then, a χ^2 failure detector is applied for attack detection. These methods rely on manually determining thresholds. Therefore, data-driven methods have gradually attracted an increasing number of attention. Utilizing a convolutional neural network-based false battery data detection and classification model, a battery data trust framework is proposed [42]. However, a large amount of reliable data is necessary for model training, and it only considers large-scale network attacks.

E. Analytical Modeling

In this part, we present a theoretical analysis of BSS status under a FDI attack and prove that the system is Finite-Time Bounder (FTB). This implies that even though the BSS system's status is compromised by a FDI attack, the stability can still be ensured through information recovery method. $\mathbb{R}^p$ refers to p -dimensional Euclidean space. $(\cdot)^T$ denotes the transpose operation. $\lambda_{\max}(\cdot)$ and $\lambda_{\min}(\cdot)$ represent the maximum and minimum eigenvalues of a matrix.

A practical BSS application scenario with S status subsystem (e.g. available, depleted and charging batteries) can be modeled by a discrete-time switched system [43], and the system state is formulated as follows:

$$x(t+1) = A_{\sigma(t)}x(t) + B_{\sigma(t)}u(t) \quad (3)$$

where $x(t) \in \mathbb{R}^p$, represents system state (e.g. the number of available batteries), $t \in \mathcal{T} \triangleq \{1, 2, \dots, T\}$ is the discrete-time instant, T denotes the number of time intervals. $\sigma(t) : [0, \infty) \rightarrow \mathcal{M} \triangleq \{1, 2, \dots, S\}$ denotes the piecewise constant switching signal. $t_j, j = 1, 2, 3, \dots$ are the switching points, and t_0 is initial time instant. If $t \in [t_j, t_{j+1})$, the $\sigma(t_j)$ th status subsystem is focused during the j -th time interval. $A_{\sigma(t)} \in \mathbb{R}^{p \times p}$ and $B_{\sigma(t)} \in \mathbb{R}^{p \times q}$ are known constant matrices, so as to upate status information. $u(t) \in \mathbb{R}^q$ is external disturbance (e.g. machine maintenance, power fluctuation), satisfying

$$\sum_{t=1}^{\infty} u^T(t)u(t) \leq \varepsilon_u^2 \quad (4)$$

where ε_u is a known positive scalar.

The battery swapping service system consists of l BSSs connected to GC, and the BSS system state information is real-time measured. At time t , the measurement equation of each BSS is given by

$$y_{\sigma(t)}^{(i)}(t) = C_{\sigma(t)}^{(i)}x(t) + D_{\sigma(t)}^{(i)}d_{\sigma(t)}^{(i)}(t), i \in \{1, \dots, l\} \quad (5)$$

where $y_{\sigma(t)}^{(i)}(t)$ is the i th BSS measurement at time instant t , $C_{\sigma(t)}^{(i)} \in \mathbb{R}^{n_d \times p}$ and $D_{\sigma(t)}^{(i)} \in \mathbb{R}^{n_d}$ are system matrices, depicting the relation between statue value and measurement value. $d_{\sigma(t)}^{(i)}(t)$ is measurement noise with $(d_{\sigma(t)}^{(i)}(t))^T d_{\sigma(t)}^{(i)}(t) \leq \varepsilon_d$, where ε_d is the boundary of measurement noises.

Equ. (5) reflects the measurement state of each BSS, but the system stability is foused from the aspect of overall network level. Therefore, we define the following global quantities, and Equ. (5) can be formulated in a compact form.

$$y_{\sigma(t)} \triangleq \left[\left(y_{\sigma(t)}^{(1)} \right)^T, \left(y_{\sigma(t)}^{(2)} \right)^T, \dots, \left(y_{\sigma(t)}^{(l)} \right)^T \right]^T \quad (6)$$

$$C_{\sigma(t)} \triangleq \left[\left(C_{\sigma(t)}^{(1)} \right)^T, \left(C_{\sigma(t)}^{(2)} \right)^T, \dots, \left(C_{\sigma(t)}^{(l)} \right)^T \right]^T \quad (7)$$

$$D_{\sigma(t)} \triangleq \text{diag} \left\{ D_{\sigma(t)}^{(1)}, D_{\sigma(t)}^{(2)}, \dots, D_{\sigma(t)}^{(l)} \right\} \quad (8)$$

$$d_{\sigma(t)} \triangleq \left[\left(d_{\sigma(t)}^{(1)} \right)^T, \left(d_{\sigma(t)}^{(2)} \right)^T, \dots, \left(d_{\sigma(t)}^{(l)} \right)^T \right]^T \quad (9)$$

On basis of above denotations, Equ. (5) thus is rewritten by

$$y_{\sigma(t)}(t) = C_{\sigma(t)}x(t) + D_{\sigma(t)}d_{\sigma(t)}(t) \quad (10)$$

Considering the vulnerabilities of BSS service system, if the status information at the i -th BSS is with FDI attack described in [44], the data received by GC could be presented as:

$$\tilde{y}_{\sigma(t)}^{(i)}(t) = f \left(y_{\sigma(t)}^{(i)}(t) \right) = \left(I + a_{\sigma(t)}^{(i)}(t) \right) y_{\sigma(t)}^{(i)}(t) \quad (11)$$

where $f \left(y_{\sigma(t)}^{(i)}(t) \right)$ is an attack function of $y_{\sigma(t)}^{(i)}(t)$, I is an identity matrix. $a_{\sigma(t)}^{(i)}(t)$ is an uncertain item, the global quantity is defined as:

$$a_{\sigma(t)} \triangleq \text{diag} \left\{ a_{\sigma(t)}^{(1)}, a_{\sigma(t)}^{(2)}, \dots, a_{\sigma(t)}^{(l)} \right\} \quad (12)$$

satisfying $a_{\sigma(t)}^T a_{\sigma(t)} \leq \varepsilon_a I$ and ε_a is a positive scalar.

Inspired by the secure state estimator based on information recovery in [45], the status data estimator at GC can be expressed:

$$\hat{x}(t+1) = R_{\sigma(t)}\hat{x}(t) + \sum_{i=1}^l W_{\sigma(t)}^{(i)} \tilde{y}_{\sigma(t)}^{(i)}(t), \quad (13)$$

where $\hat{x}(t)$ is the state estimate, $R_{\sigma(t)}$ and $W_{\sigma(t)}$ are weight coefficients.

Let $e(t) = x(t) - \hat{x}(t)$ be estimation error, we can obtain estimation error system as:

$$\alpha(t+1) = \tilde{A}_{\sigma(t)}\alpha(t) + \tilde{B}_{\sigma(t)}\gamma(t) \quad (14)$$

where $\alpha(t) = \begin{bmatrix} x(t) \\ e(t) \end{bmatrix}$, $\gamma(t) = \begin{bmatrix} u(t) \\ d_{\sigma(t)}(t) \end{bmatrix}$

$$\tilde{A}_{\sigma(t)} = \begin{bmatrix} A_{\sigma(t)} & A_{\sigma(t)} & O \\ A_{\sigma(t)} - R_{\sigma(t)} - W_{\sigma(t)}(I + a_{\sigma(t)})C_{\sigma(t)} & R_{\sigma(t)} \end{bmatrix}$$

$$\tilde{B}_{\sigma(t)} = \begin{bmatrix} B_{\sigma(t)} & O \\ B_{\sigma(t)} & -W_{\sigma(t)}(I + a_{\sigma(t)})D_{\sigma(t)} \end{bmatrix}$$

Definition 1 (*Finite-Time Bounded, FTB [46]*). Given a positive definite matrix R and $c_2 > c_1$, the secure estimation error system (12) is said to be FTB with respect to $(c_1, c_2, R, T, \varepsilon_u)$, if $\alpha^T(t_0) R \alpha(t_0) \leq c_1 \Rightarrow \alpha^T(t) R \alpha(t) \leq c_2$

Proof. We choose the quadratic Lyapunov function $V_k(t)$ as following:

$$V_k(t) = \alpha^T(t)G_k\alpha(t), t \in [t_s, t_{s+1}), k \in \mathcal{M} \quad (15)$$

where G_k is a positive definite matrix.

Equ. (14) presents the expression of estimation error system, substituting it to Equ. (15), we can get the value of $V_k(t+1)$:

$$V_k(t+1) = \begin{pmatrix} \alpha(t) \\ \gamma(t) \end{pmatrix}^T \begin{pmatrix} \tilde{A}_k^T G_k \tilde{A}_k & \tilde{A}_k^T G_k \tilde{B}_k \\ \tilde{B}_k^T G_k \tilde{A}_k & \tilde{B}_k^T G_k \tilde{B}_k \end{pmatrix} \begin{pmatrix} \alpha(t) \\ \gamma(t) \end{pmatrix} \quad (16)$$

According to the following constrain:

$$\begin{pmatrix} \tilde{A}_k^T G_k \tilde{A}_k & \tilde{A}_k^T G_k \tilde{B}_k \\ \tilde{B}_k^T G_k \tilde{A}_k & \tilde{B}_k^T G_k \tilde{B}_k \end{pmatrix} < \begin{pmatrix} \kappa G_k & O \\ O & \delta^2 I \end{pmatrix} \quad (17)$$

where $\kappa \geq 1, \delta > 0$ are two scalars. Therefore, Equ. (16) is derived as:

$$\begin{aligned} V_k(t+1) &\leq \kappa \alpha^T(t) G_k \alpha(t) + \delta^2 \gamma^T(t) \gamma(t) \\ &= \kappa V_k(t) + \delta^2 \gamma^T(t) \gamma(t). \end{aligned} \quad (18)$$

On basis of Equ. (18), the following inequality is obtained:

$$V_{\sigma(t)}(t) \leq \kappa^{t-t_s} V_{\sigma(t_s)}(t_s) + \delta^2 \sum_{j=t_s}^{t-1} \kappa^{t-j} \gamma^T(j) \gamma(j)$$

$$\begin{aligned} &\leq \kappa^{t-t_0} \xi^{\frac{(t-t_0)}{\tau}} V_{\sigma(t_0)}(t_0) + \delta^2 \kappa^t (\varepsilon_u^2 + T\varepsilon_d^2) \quad (19) \\ &+ \xi \delta^2 (\varepsilon_u^2 + T\varepsilon_d^2) \sum_{s=1}^{\frac{(t-t_0)}{\tau}} \kappa^{t-t_s} \xi^{\frac{(t-t_0)}{\tau}-s} \end{aligned}$$

where $\xi > 1$ is a scalar and $\frac{\tau - t}{\tau}$ is the average dwell time for switching signal $\sigma(t)$. Note that t_0 is the initial time instant, let $\sigma(t) \rightarrow k$, $t_0 \rightarrow t$, Equ. (15) is derived as:

$$V_{\sigma(t_0)}(t_0) = \alpha^T(t_0)G_{\sigma(t_0)}\alpha(t_0) \quad (20)$$

Then, we substitute Equ.(20) into (19)

$$V_{\sigma(t)}(t) \leq \kappa^T \delta^{\frac{T}{\tau}} c_1 \lambda_1 + \delta^2 \kappa^T (\varepsilon_u^2 + T \varepsilon_d^2) + \xi \delta^2 (\varepsilon_u^2 + T \varepsilon_d^2) \quad (21)$$

where $\lambda_1 = \max \left\{ \lambda_{\max}(R^{-\frac{1}{2}} G_k R^{-\frac{1}{2}}) | k \in \mathcal{M} \right\}$ and R is positive definite matrix. Besides, we notice that

$$V_{\sigma(t)}(t) = \alpha^T(t)G_{\sigma(t)}\alpha(t) \geq \lambda_2\alpha^T(t)R\alpha(t) \quad (22)$$

$$\text{where } \lambda_2 = \min \left\{ \lambda_{\min}(R^{-\frac{1}{2}} G_k R^{-\frac{1}{2}}) | k \in \mathcal{M} \right\}.$$

With the result of $V_{\sigma(t)}(t)$ in Equ. (21), the value of $\alpha^T(t)R\alpha(t)$ in Equ. (22) right-side is further derived:

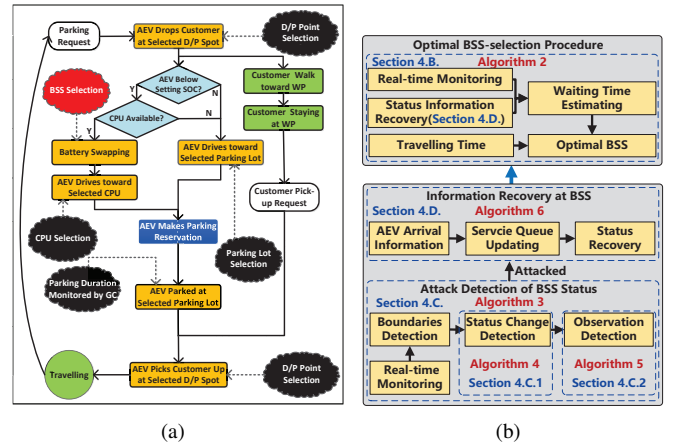


Fig. 4. The system operation logic. (a) Operation logic of on-the-move AEV battery swapping and parking charging decision-making. (b) The optimal BSS-selection decision-making under FDI attack.

$$\begin{aligned} \alpha^T(t)R\alpha(t) &\leq \frac{1}{\lambda_2}[\kappa^T\delta^{\frac{T}{\tau}}c_1\lambda_1 + \delta^2\kappa^T(\varepsilon_u^2 + T\varepsilon_d^2) \\ &\quad + \xi\delta^2(\varepsilon_u^2 + T\varepsilon_d^2)] \triangleq c_2 \end{aligned} \quad (23)$$

Remark. According to Definition 1, the estimation error system is proved to be finite-time bounded. As claimed in [46], the main difference between FTB and Finite-Time Stabilizable (FTS) is whether the disturbance is considered in system Equ. (3), and the verification of FTB is equivalent to prove the system is stable. Therefore, despite potential attacks on status of BSS, the service stability can still be maintained.

IV. SYSTEM DESIGN

A. Parking and Energy Supplement Decision-making

Fig. 4(a) reveals the decision-making logic about parking and energy supplement. Meanwhile, the detailed implement procedure is depicted in Alg. 1. The optimal D/P point l_{dp} is obtained by following formulation:

$$\arg \min_{l_{dp} \in \Lambda_{dp}} T_{cx}^{trip(l_{dp})} \quad (24)$$

s.t.,

$$T_{cx}^{trip(l_{dp})} = T_{cx}^{aev, l_{dp}} + T_{cx}^{l_{dp}, wp} \quad (25)$$

where Λ_{dp} is the set of locations of D/P. $T_{cx}^{trip(l_{dp})}$ represents outbound (or inbound) trip duration for customer, consisting of $T_{cx}^{aev, l_{dp}}$ (AEV traveling duration from current location to D/P point l_{dp}) and $T_{cx}^{l_{dp}, wp}$ (customer walking time from D/P point to trip destination). The objective function jointly minimizes AEV traveling time and users' satisfaction (walking time) through a desirable commuter bus pick-up point, to decouple the LAVP into autonomous parking and user motion towards destination. Through the designed LAVP service strategy based on above objective function, the optimal D/P point is determined with the service characteristic of a commuter bus to guarantee users' satisfaction.

After dropping off customers at the selected D/P point l_{dp} , if the AEV's SOC value is below the threshold, once any CPUs are expected to be available after battery swapping operation, the AEV prefers battery swapping service. This integration of BSS is advantageous for AEVs parking at CPUs, ensuring rapid and complete energy replenishment. The optimal BSS l_{hss}^{opt} is obtained by Alg. 2. Although Alg. 1 does not directly

Algorithm 1 Decision-making on Energy Supplement and Parking

```

1: for each AEV after drop-off phase do
2:   if AEV's battery level is below the SOC threshold then
3:     if CPU is available then
4:       determine the optimal BSS  $l_{bss}^{opt}$  via Algorithm 2
5:       move towards and make reservation at  $l_{bss}^{opt}$ 
6:       after accomplishing battery swapping operation for AEV
7:       recommend the optimal CPU  $l_{cpu}^{opt}$  to AEV
8:     else
9:       recommend the optimal CPR  $l_{cpr}^{opt}$  to AEV
10:    end if
11:  else
12:    determine the optimal parking lot  $l_{pl}^{opt}$  among CPU and CPR
13:    make parking reservation at  $l_{pl}^{opt}$ 
14:    AEV move towards and parked at  $l_{pl}^{opt}$ 
15:  end if
16: end for

```

consider the waiting time for BSS during peak demand, Alg. 2 is nested in Alg. 1 and comprehensively considers how to calculate the waiting time for BSS. Subsequently, the AEV drivers toward and make reservation at l_{bss}^{opt} (line 3 to 5 in Alg. 1). After battery swapping operation, the optimal CPU l_{cpu}^{opt} is thus recommended to AEV according to following equation:

$$\arg \min_{l_{cpu}^{opt} \in \Lambda_{cpu}} \left(T_{aev}^{l_{dp},cpu} + EW_{aev}^{cpu} \right) \quad (26)$$

where $T_{aev}^{l_{dp},cpu}$ is traveling time from D/P point to CPU. EW_{aev}^{cpu} represents the waiting time until a parking slot at CPU becomes available. Otherwise, if all CPUs are occupied, GC will recommend an optimal CPR to an AEV, via minimizing the summation of charging waiting time and traveling time from D/P point to CPR (line 6 to 9 in Alg. 1). The optimal CPR is also obtained with the same formulation of Equ. (26), and the only difference is the variate is related to CPR rather than CPU.

After dropping off the customer, if the SOC value of AEV is above threshold, the focus shifts solely to parking decision-making. In this scenario, the charging functionality of CPRs is not a concern, and emphasis is placed on the convenience and availability of parking. The AEV then proceeds to make a parking reservation and navigate towards the optimal parking lot l_{pl}^{opt} (line 11 to 15 in Alg. 1). The optimal parking slot l_{pl}^{opt} is also obtained based on Equ. (26). In this procedure, the optimal parking lot is determined among all CPUs and CPRs.

B. The Optimal BSS-selection Procedure

Based on the OCPP communication protocol as introduced in Section III-C, the real-time monitoring data of BSS local status can be collected at GC for optimal BSS recommendation. The methodology for this process, including how to identify and mitigate the impact of FDI attacks on BSS selection, is detailed in Alg. 2 and illustrated in Fig. 4(b). Define the total number of BSSs over network is n , the total number of charging slots is m , and the total number of AEVs is k . The time complexity of optimal BSS-selection decision-making as depicted in Fig. 4(b) is $O(km^2n)$.

In this manuscript, the current status of BSS is real-time obtained via a communication network. For the future service status estimation of BSS, the technique based on heuristic method via list update and sort is used. To be specific,

according to current local status information of BSS, the available time of all batteries at BSS can be calculated. In reservation queue, AEVs arriving earlier than the target AEV have a higher priority service order, and the number of available batteries at BSS will decrease. Meanwhile, the removed batteries from AEVs that arrived earlier than the target AEV will also participate in BSS status update. To sum up, the manuscript utilizes interactive update mechanism of double queue information (i.e. BSS local status queue and global reservation information queue) to determine the status of BSS. The estimation technique used in this manuscript to determine the status of BSS is introduced detailedly in the following paragraphs.

In scenarios where BSS data may be tampered due to FDI attacks, Alg. 3 ascertains the reliability of the BSS's real-time monitoring data. If $attackFlag = false$, it indicates the absence of an attack, and hence, the local status information is regarded as reliable for optimal BSS selection. Conversely, $attackFlag = true$ suggests a successful FDI attack, enabling the replacement of current local status data with recovery values, as computed by Alg. 6 (line 3 to 8 in Alg. 2).

The battery charging duration T^{cha} is calculated by:

$$T^{cha} = (E^{max} - E^{cur})/P_0 \quad (27)$$

where E^{max} is the maximum capacity of battery, E^{cur} is the current battery capacity and P_0 is the rated charging power.

Algorithm 2 The optimal BSS-selection based on local status

```

1: for each BSS over service network do
2:   receive BSS local status  $n_B, N_C, N_D$  via real-time monitoring
3:   get the value of  $attackFlag$  return by Algorithm 3
4:   if  $attackFlag = false$  then
5:     utilizing real-time sensing data  $n_B, N_C, N_D$ 
6:   else
7:     replace tampering data with status recovery via Algorithm 6
8:   end if
9:   for each charging battery in  $N_C$  and waiting for charging in  $N_D$  do
10:    record completing charging time in  $TList$ 
11:    if reservation list  $N_R$  is not empty then
12:      for each AEV in  $N_R$  expected arrives earlier than  $t^{arr}$  do
13:         $n_B = n_B - 1$  and refine  $TList$ 
14:      end for
15:    end if
16:  end for
17:  prioritize  $TList$  with ascending order
18:   $T^{tot} = EWTS + T^{tra}$ 
19: end for
20:  $l_{bss}^{opt} \leftarrow \arg \min(T^{tot})$ 

```

The charging finishing time t^{fin} is then derived as:

$$t^{fin} = t^{cur} + T^{cha} \quad (28)$$

where t^{cur} represents the current time. For all batteries being charged and to be charged, the expected charging finishing time t^{fin} is recored at list $TList$. The reservation list N_R includes expected arrival time t^{cur} and remain energy for upcoming AEVs. For an target AEV with the requirement of battery swapping service, if there are any other AEVs in N_R arrive earlier, it means these AEVs can experience a higher service priority. This is because the service order at BSS follows First Come First Serve (FCFS) rule. Therefore, the number of expected available batteries n_B decreases by one. Meanwhile, the removed battery from AEVs in N_R is utilized to refine $TList$. Eventually, the $TList$ is updated and sorted

according to ascending order (line 9 to 17 in Alg. 2).

Based on Equ. (2), the value of $EWTS$ can be obtained. Then, the total time cost T^{tot} is obtained by the summation of T^{tra} (the traveling time from current location to BSS) and $EWTS$. Consequently, the BSS with minimum time cost is recommended (line 18 to 20 in Alg. 2).

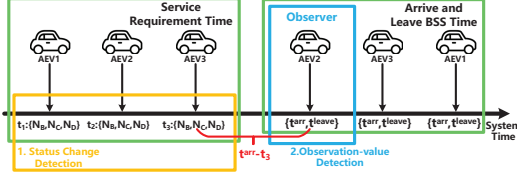


Fig. 5. The illustration of attack detection about BSS local status data

C. The Attack Detection at BSS

To effectively detect FDI attacks at BSS, we propose a status detection mechanism outlined in Alg. 3. This mechanism encompasses two primary steps: status change detection (Alg. 4) and observation-value detection (Alg. 5). The overall detection process is visually represented in Fig. 5.

In the first phase of local status detection, the variable $attackFlag$ is initialized as false to indicate the absence of an attack. Spanning from line 2 to 11 in Alg. 3, when an AEV requests battery swapping service, the GC gathers real-time monitoring data from the BSS. Then, the data boundary of status information is verified, and the current demand time t_{cur} is recorded in RLIST. Corresponding sensing status information, including n_B, N_C, N_D , is stored in BStatusHashmap, CStatusHashmap, and DStatusHashmap, respectively.

The second phase, detailed from line 12 to line 30 in Alg. 3, focuses on the AEV's arrival process at BSS l at time t^{arr} . Upon arrival, the size of RLIST is checked. If $RLIST.size()$ is non-zero, indicating that the AEV is the earliest to arrive, $checkAEVFlag$ is set to true for subsequent observation-value detection. RLIST is then sorted in ascending order to reflect the timeline of service requirements. For multiple service requests ($RLIST.size() > 1$), local status changes for each time slot are detected using Alg. 4, with the largest time value in RLIST denoted as t_0 (eg: t_3 in Fig. 5). The number of AEVs awaiting service before AEV i (n_{bef}^{arr}) is calculated:

$$n_{bef}^{arr} = n_l^{arr} - n_l^{lea} - 1 \quad (29)$$

where n_l^{arr} is the number of AEVs arriving at BSS l and n_l^{lea} is the number of AEVs leaving. Since the charging power at BSS is constant, based on the queue information of N_C, N_D , the number of batteries finishes charging to be available from time t_0 to t_{arr} is defined as n^{fc} . If $n_B + n^{fc} > n_{bef}^{arr}$, immediate battery swapping is possible. The expected waiting time $\hat{T}^{wait}$ is then the operation time for battery swapping (ρ^{swap}). Otherwise, since there are n_{bef}^{arr} AEVs waiting for service before AEV i , the service order of AEV i is $n_{bef}^{arr} + 1$. Namely, when $n_{bef}^{arr} + 1$ -th battery is fully charged at time t^{fin} , it is the turn for target AEV.

Accordingly, $\hat{T}^{wait} = t^{fin} - t^{arr} + \rho^{swap}$. After above processing, all elements in RLIST are removed and RLIST is set with null. Finally, as depicted at line 31, the third phase verifies expected waiting time $\hat{T}^{wait}$ with actual waiting time. The detailed procedure is implemented in Alg. 5.

Algorithm 3 A detection mechanism of BSS status

```

1: define  $attackFlag = false$ ,  $checkAEVFlag = false$ 
2: if an AEV requires battery swapping service then
3:   obtain BSS local status  $n_B, N_C, N_D$  via real-time monitoring
4:   if  $(n_B > N^0) \vee (|N_C| > \theta^0) \vee (n_B + |N_C| + |N_D| \neq N^0)$  then
5:     return  $attackFlag = true$ 
6:   else
7:     RLIST.ADD( $t_{cur}$ ), BStatusHashmap.ADD( $t_{cur}, n_B$ )
8:     CStatusHashmap.ADD( $t_{cur}, N_C$ )
9:     DStatusHashmap.ADD( $t_{cur}, N_D$ )
10:  end if
11: end if
12: when an AEV  $i$  arrives BSS  $l$  at time  $t^{arr}$ 
13: if  $(RLIST.size() \neq 0)$  then
14:    $checkAEVFlag = true$ 
15:   sort RLIST according to ascending order
16:   if  $(RLIST.size() > 1)$  then
17:     detect sensing status change via Algorithm 4
18:   end if
19:   obtain the largest time instant value  $t^0$  in RLIST
20:   count number of AEVs waiting for service before AEV  $i$  as  $n_{bef}^{arr}$ 
21:   obtain sensing BSS local status  $n_B, N_C, N_D$  at  $t^0$ 
22:   calculate number of batteries ( $n^{fc}$ ) finishing charging from time  $t^0$ 
   to  $t^{arr}$  via  $N_C, N_D$ 
23:   if  $(n_B + n^{fc} > n_{bef}^{arr})$  then
24:      $\hat{T}^{wait} = \rho^{swap}$ 
25:   else
26:     calculate  $(n_{bef}^{arr} + 1) - th$  battery charging finishing time via
        $N_C, N_D$  as  $t^{fin}$ 
27:      $\hat{T}^{wait} = t^{fin} - t^{arr} + \rho^{swap}$ 
28:   end if
29:   remove all elements of RLIST
30: end if
31: observation-value detection via Algorithm 5

```

Algorithm 4 Status change detection

```

1: receive RLIST, BStatusHashmap, CStatusHashmap, DStatusHashmap
   from Algorithm 3; initialize ACLIST, FTList and NDList as null
2: for  $(i = 1; i \leq RLIST.size() - 1; i++)$  do
3:    $t^1 = RLIST.GET(i)$ 
4:    $t^2 = RLIST.GET(i + 1)$ 
5:    $T^{dif} = t^2 - t^1$ 
6:   obtain  $n_B^1, n_B^2$  at  $t^1, t^2$  via BStatusHashmap
7:   obtain  $N_C^1, N_C^2$  at  $t^1, t^2$  via CStatusHashmap
8:   obtain  $N_D^1, N_D^2$  at  $t^1, t^2$  via DStatusHashmap
9:   for  $(i = 1; i \leq |N_C^1|; i++)$  do
10:    if  $(T_i^{cha} > T^{dif})$  then
11:      ACLIST.ADD( $E^{update}$ )
12:    else
13:       $n_B^1++$ 
14:      ACLIST.remove( $E^i$ ), FTList.ADD( $t_1 + T_i^{cha}$ )
15:    end if
16:  end for
17: while  $(ACLIST.size() < \theta) \&\& (N_D \neq null)$  do
18:    $n_s = \theta - ACLIST.size()$ 
19:   remove first  $n_s$  elements of  $N_D$  into NDList based on STCF
20:   for  $(k = 1; k \leq NDList.size(); k++)$  do
21:     sort FTList according to ascending order
22:     if  $(T_k^{cha} + FTList.GET(k) > T^{dif})$  then
23:       ACLIST.ADD( $E^{update}$ )
24:     else
25:        $n_B^1++$ 
26:       replace  $T_k^{cha}$  in FTList with  $(T_k^{cha} + FTList.GET(k))$ 
27:     end if
28:   end for
29: end while
30: end for
31: if  $(n_B^1 == n_B^2) \&\& (ACLIST.equal(N_C^2))$  then
32:   return  $attackFlag = false$ 
33: else
34:   return  $attackFlag = true$ 
35: end if

```

1) *Real-time Monitoring Status Detection*: As depicted from line 1 to 2 in Alg. 4, the requirement time and local status lists are obtained according to Alg. 3. For each service request time in RLIST, adjacent times are defined as t^1 and t^2 respectively. The time interval T^{dif} between these moments is calculated (line 5 in Alg. 4). Subsequently, the BSS's local status at these times is determined (line 6 to 8 in Alg. 4).

The expected charging time (T_i^{cha}) for each battery in the charging queue N_C^1 at time t^1 is calculated according to Equ. (27). If $T_i^{cha} > T^{dif}$, the battery cannot be fully charged within time interval T^{dif} . The updated battery energy E^{update} after this interval is thus derived by

$$E^{update} = E_i^{cur} + P_0 \cdot T^{dif} \quad (30)$$

Otherwise, the battery will be fully charged, increasing the count of available batteries. The moment of charging finishing time ($t_1 + T_i^{cha}$) is appended into FTList. The time when charging finishes, ($t_1 + T_i^{cha}$), is added to FTList, which is then sorted in ascending order (line 9 to 16 in Alg. 4).

If $(ACList.size() < \theta) \&\& (N_D \neq null)$, it means that there are some vacant charging slots and depleted batteries waiting for charging. Thus, the recharging process of depleted battery is our concern (line 17 to 19 in Alg. 4). The NDList records batteries in N_D will be charged during time interval T^{dif} . For each battery in NDList, the energy update strategy is with similar idea of batteries being charged in N_C^1 (line 20 to 28 in Alg. 4). Finally, the calculated local status values for time t_2 are derived based on queue information from time t_1 . As indicated in line 31 to 35, these calculated values are compared with the actual values at time t_2 . Inconsistencies indicate a potential FDI attack at the BSS, thereby setting the *attackFlag* to true.

Algorithm 5 Observation-value detection

```

1: receive  $\hat{T}^{wait}$  from Algorithm 3
2: when an AEV arrives BSS  $l$  at  $t^{lea}$ 
3: if (checkAEVFlag = true) then
4:    $T^{wait} = t^{lea} - t^{arr}$ 
5:   if ( $T^{wait} \neq \hat{T}^{wait}$ ) then
6:     return attackFlag = true
7:   end if
8: end if
```

2) *Observation-value Detection*: This part focuses on evaluating the reliability of local status via observation-value, as presented in Alg. 5. When an AEV is flagged with *checkAEVFlag* = true, it assumes the role of an “observer”, as depicted in Fig. 5. The observer AEV's expected waiting time, $\hat{T}^{wait}$, is estimated in accordance with the provisions of Alg. 3. Subsequently, the actual waiting time for this observer AEV, denoted as T^{wait} , is calculated using the departure time t^{lea} and the arrival time t^{arr} .

A difference between the estimated waiting time $\hat{T}^{wait}$ and the actual waiting time T^{wait} represents potential inaccuracies in the local status. Such inconsistencies suggest that the BSS may be under the influence of an attack. Consequently, the *attackFlag* is set to true, indicating a unreliable status of the BSS and requiring further corrective measures.

D. Information Recovery at BSS

As shown at line 1 in Alg. 6, the number of available batteries at time t is defined as $\tilde{n}_B(t)$, and initialized as the number of initial stock battery N_B^0 at BSS. Similarly, the available charging slots at time t , represented as $\tilde{\theta}(t)$, start with the initial number of charging slots θ^0 at the BSS. Additionally, $\tilde{N}_C$ and $\tilde{N}_D$ track the queues of batteries being charged and those that are depleted, respectively, while $\tilde{N}_W$ reflects the queue of AEVs waiting for service.

Algorithm 6 Information recovery at BSSs

```

1: define  $\tilde{n}_B(t) = N_B^0, \tilde{\theta}(t) = \theta^0, \tilde{N}_C, \tilde{N}_D, \tilde{N}_W = null$ 
2: when an AEV  $i$  arriving at time  $t_i^{arr}$  with energy  $E_i^{cur}$ 
3: if  $\tilde{n}_B > 0$  then
4:    $t_i^{swap} = t_i^{arr} + \rho^{swap}$ 
5:    $\tilde{n}_B(t) = \tilde{n}_B(t) - \varepsilon(t - t_i^{swap})$ 
6:   if  $\tilde{\theta}(t_i^{swap}) > 0$  then
7:     add  $\langle t_i^{swap}, E_i^{cur} \rangle$  into ChaHashmap
8:      $\tilde{\theta}(t) = \tilde{\theta}(t) - \varepsilon(t - t_i^{swap})$ 
9:      $t_i^{fin} = t_i^{swap} + T_i^{cha}$ 
10:     $\tilde{n}_B(t) = \tilde{n}_B(t) + \varepsilon(t - t_i^{fin})$ 
11:     $\tilde{\theta}(t) = \tilde{\theta}(t) + \varepsilon(t - t_i^{fin})$ 
12:   else
13:     add  $\langle t_i^{swap}, E_i^{cur} \rangle$  into DepHashmap
14:   end if
15: else
16:   add AEV  $i$  into  $\tilde{N}_W$  and prioritize  $\tilde{N}_W$  according to FCFS
17: end if
18: if current time  $t^{cur}$  in ChaHashmap then
19:   add value of ChaHashmap( $t^{cur}$ ) into  $\tilde{N}_C$ 
20: end if
21: if current time  $t^{cur}$  in DepHashmap then
22:   add value of DepHashmap( $t^{cur}$ ) into  $\tilde{N}_D$ 
23: end if
24: define  $\hat{N}_C = \tilde{N}_C, \hat{N}_D = \tilde{N}_D$ 
25: for ( $j = 1; j \leq \tilde{N}_C; j++$ ) do
26:   update  $\hat{N}_C$  via  $E_j^{cur} = E_j^{cur} + \tau \cdot P_0$ 
27:   if  $E_j^{cur} \geq E^{max}$  then
28:     remove battery  $j$  from  $\hat{N}_C$ 
29:   if  $\tilde{N}_W$  is not empty then
30:      $t_i^{swap} = t^{cur}$ , execute battery swapping as line 5 to 13
31:   end if
32:   if  $\tilde{N}_D$  is not empty then
33:     sort  $\tilde{N}_D$  according to STCF
34:     remove the first element of  $\tilde{N}_D$  into  $\tilde{N}_C$ 
35:   end if
36: end if
37: end for
38: return  $\tilde{n}_B(t^{cur}), \hat{N}_C, \hat{N}_D$ 
```

If there are available batteries at BSS when an AEV arrives, the AEV experiences battery swapping service immediately. The operation, lasting ρ^{swap} , completes at t_i^{swap} , and subsequently the count of available batteries is reduced. The step-up function $\varepsilon(\cdot)$ helps to depict the time-dependent distribution of batteries. After battery swapping operation, if there are available charging slots, the depleted battery will be charged immediately (from line 6 to 12 in Alg. 6).

As for status queue update, the current energy of depleted battery will be recorded in charging list $\tilde{N}_C$. Meanwhile, the moment t_i^{swap} and corresponding list $\tilde{N}_C$ are incorporated as key-value, added into *ChaHashmap*. The number of available batteries accordingly decrease by one at time t_i^{swap} . t_i^{fin} is the time of depleted battery finishing charging. At time t_i^{fin} , the number of available batteries increase and occupied charging slot is released. By contrast, if the depleted battery

removed from AEV while all charging slots are occupied, the battery will be added into depleted battery queue $\tilde{N}_D$ waiting for recharging. $DepHashmap$ records the time t_i^{swap} and depleted battery queue $\tilde{N}_D$. In another scenario, there are no enough batteries when an AEV i arrives at BSS, the AEV will be appended into queue $\tilde{N}_W$ to wait charging battery becoming available (from line 13 to 17 in Alg. 6). As depicted from line 18 to 23 in Alg. 6, if current time t^{cur} is the key in $ChaHashmap$ or $DepHashmap$, the corresponding value is added into $\tilde{N}_C$ or $\tilde{N}_D$. Thus, the local status queue information can be obtained. The system's status update for next time interval is depicted from line 24 to 36. Finally, This process obtains the estimation of local status information ($\tilde{n}_B(t^{cur})$, $\tilde{N}_C$, $\tilde{N}_D$), and is then utilized for information recovery.

With the development of cyber-attack based on AI, a novelty dynamic FDI attack method is developed via smart Q-learning algorithm to implement optimal attack strategy from actions and gains [47]. Facing such an intelligent and dynamic attack method, the proposed detection framework in this paper can still be effective. This is because the framework is referred as behavior-based detection mechanism. In this proposed framework, although dynamic FDI attacks that identify vulnerabilities in static defence procedures and inject bounded, stealth manipulations to bypass them, the detection mechanism can still check the sensing status information reliability based on sensing data correlation. Besides, even if all sensing data is tampered to satisfy status change detection, the actual service waiting time of "Observer" (an AEV with $checkAEVFlag = true$) in Algorithm 5 will be compared to a calculated value based on sensing data. Therefore, so long as BSS status information is tampered, the attack can be detected.

implemented through developing the routing part of ONE. The GUI of real-time simulation platform is presented in Fig. 6(b).

In Fig. 7, the default simulation scenario is abstracted from Helsinki city road topology in Finland with $4500 \times 3400 m^2$ area. According to the practical traffic flow situation, an initial number of 300 AEVs are with a fluctuating speed of $[30 \sim 50] km/h$ in the network. All AEVs are configured based on the energy specification Maximum Capacity, Max Traveling Distance, SOC threshold. The energy configuration is set with $\{16.4 kWh, 140 km, 50\% \}$ according to Hyundai BlueOn [49]. For simulation purposes, AEV drop-off requests are scheduled at 3600s, with a subsequent parking duration of 7200s. The simulation spans 12 hours, with a network update interval of $\tau = 0.1$ seconds. All AEVs are configured based on the energy specification $\{Maximum Capacity, Max Traveling Distance, SOC threshold\}$. AEVs are navigated to their designated points using the shortest route based on the road topology.

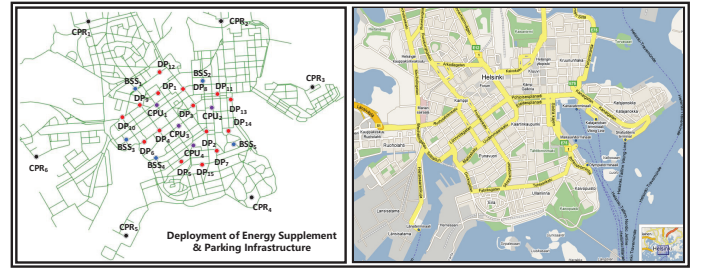


Fig. 7. Default simulation scenario of Helsinki city

The infrastructure for parking and energy supplementation includes 6 CPRs with ample parking space for all vehicles. Each CPR offers 30 charging slots with a capacity of 10 kW. The city map incorporates 15 D/P points, supplemented by 5 downtown-located BSSs and 4 CPUs. Based on the definition of CPU in Section III-A, the CPU is modeled as a mechanical parking garage. Each BSS is configured with 30 battery stocks and 30 charging slots rated at 60 kW to recharge depleted batteries in parallel. Due to each AEVs at mechanical parking garage is automatically parked via lifting and movement device, AEVs is unable to keep a stable wired link to charging pile. Therefore, CPU cannot provide charging services based on wired connection for AEVs. Additionally, each CPU is configured with 20 parking slots for AEVs.

The following schemes are conducted for comparison:

- **LAVP-CHARGE-DST (LCD):** The optimal CPR is obtained according to the shortest distance between target D/P point and CPRs, according to detail in [18].
- **LAVP-CHARGE-RES (LCR):** The optimal CPR selection is based on the minimum expected waiting time for charging and the distance from the target D/P point, according to detail in [6].
- **LAVP-HYBRID-RES (LHR):** The proposed LAVP and energy supplement service strategy. Meanwhile, the CPU is also incorporated to improve the parking convenience. (According to Algorithm 2)
- **LAVP-HYBRID-LOCAL (LHL):** Similar to LHR, but decision-making about parking and energy supplementation relies solely on local status. (Based on Algorithm 2,

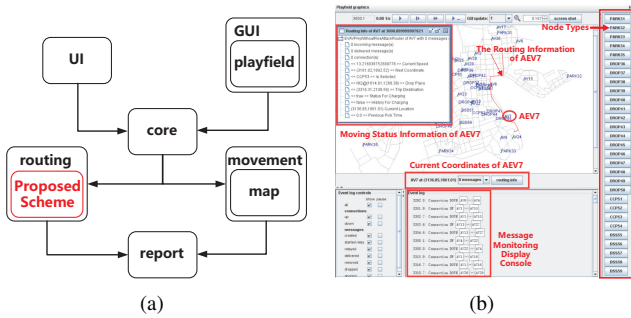


Fig. 6. The introduction of ONE system. (a) architecture. (b) GUI.

V. PERFORMANCE EVALUATION

A. Simulation Configuration

The real-time simulation platform utilizes the Opportunistic Network Environment (ONE) simulator for analysis [48], and the simulator is an open-source software based on Java. The whole system architecture of ONE is depicted in Fig.6(a). The core components of ONE, such as the host and its classes for inter-host connections, are all included in the core package. Throughout the simulation process, the routing module and mobile module provide data sources for the report module. Based on the ONE, we have carried out a secondary development to integrate service system for AEV energy supplementation and LAVP. The detailed proposed scheme is

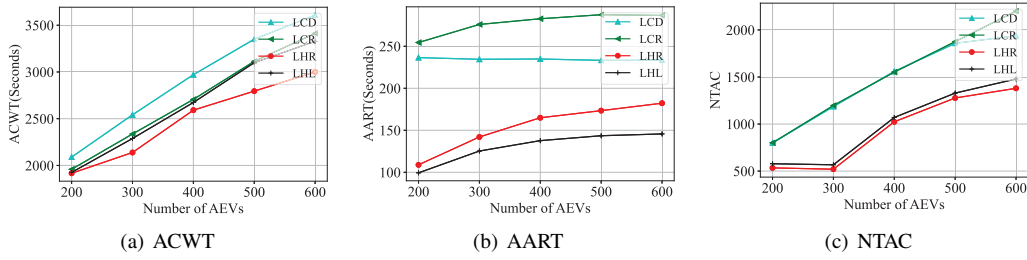


Fig. 8. Service optimization: Impact of AEVs density

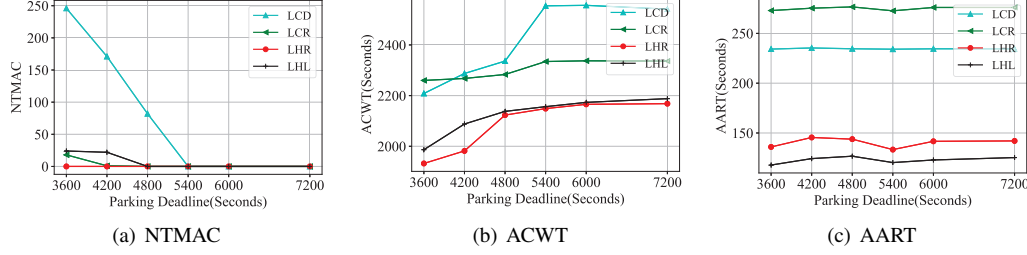


Fig. 9. Service optimization: Impact of parking deadline

excluding the reservation step)

Meanwhile, the following metrics are adopted for performance evaluation:

- **Average Charging Waiting Time (ACWT):** The average duration an AEV spends at the selected CPR, lasting from the time it arrives until finishing charging.
- **Average AEV Response Time (AART):** The average travel duration for an AEV from the parking slot to the D/P point.
- **Number of Total AEV Charging (NTAC):** The total number of AEVs charged at all CPRs. It reflects the charging busyness of CPR.
- **Number of Missed AEV Charging (NTMAC):** The total number of AEVs that be charged but not reach the full of battery capacity until they leaves.

B. Scenario I: Research on Energy Supplement and LAVP Service Optimization

In this subsection, we examine the effect of integrating service optimization for energy supplement with LAVP. The evaluation begins by simulating scenarios without FDI attacks. Based on these analyses, simulation results under FDI attack will further be depicted in Section V-C.

1) *Impact of AEV Density:* As depicted in Fig. 8(a), the proposed LHR scheme achieves the lowest ACWT. This is attributed to the LHR scheme taking advantage of BSS and reservation mechanism, where AEVs can be fully refilled in a short period and future demand can be predicated, thus alleviating charging service congestion at CPR. It is observed that for the LHR scheme, the growth trend of ACWT becomes more remarkable as the number of AEVs increases from 300 to 400. This is rationalized by the fact that the selection of battery swapping service is dependent on the availability of CPU. Once all CPUs are occupied, the AEV has to move towards CPRs for parking and charging, thereby increasing service response time. By utilizing the advantage of CPUs, the proposed LHR and LHL schemes exhibit lower AART as shown in Fig. 8(b). Moreover, in an effort to balance service

load across the network, LHR might recommend a parking slot that are farther away to WP, resulting in a longer AART compared to LHL. Additionally, NTAC shows a significant increase once the number of AEVs exceeds 300 as illustrated in Fig. 8(c), highlighting the BSS's effectiveness in alleviating CPR charging pressure.

2) *Impact of Parking Deadline:* The parking deadline, denoting the maximum time an AEV may occupy a parking slot, significantly influences service effectiveness, as depicted in Fig. 9. Fig. 9(a) shows that when facing a short parking deadline, such as 3600 seconds, a notable number of AEVs are unable to fully charge, particularly under the LCD scheme. The LCD scheme focuses on distance, while neglecting waiting times, thus causing local congestion. Consequently, under restricted deadline constraints, AEVs are compelled to depart CPRs without full charging, leading to an increase in the NTMAC. Hence, we have configured the system simulation with a parking duration of 7200 seconds, ensuring that all AEVs are fully charged at CPRs. Notably, the proposed LHR scheme demonstrates superior performance compared to other algorithms, successfully fulfilling the charging needs of all AEVs even under shorter parking deadlines and thereby enhancing the efficiency of LAVP services, as depicted in Fig. 9(b) and Fig. 8(b).

C. Scenario II: Research on FDI Attack

To further investigate the impact of FDI attack and verify the effectiveness of proposed method, the following schemes are included for comparison:

- **Attack-LAVP-HYBRID-RES (ALHR):** Unlike LHR, the ALHR scheme assumes the status information to be reliable and does not factor in the possibility of an attack.
- **Attack-LAVP-HYBRID-LOCAL (ALHL):** Different from ALHR, the ALHL scheme does not consider service reservation information, instead of selecting BSS and parking slot according to local service status.

Apart from the CPR side, the performance metrics of ACWT and NTAC can also be defined at BSS side, offering

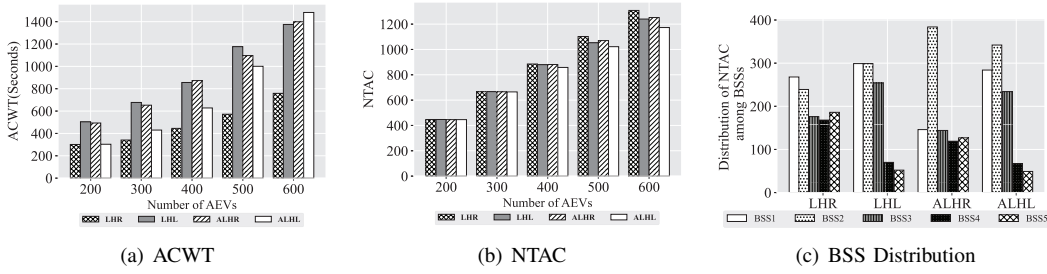


Fig. 10. FDI attack: Impact of AEVs density

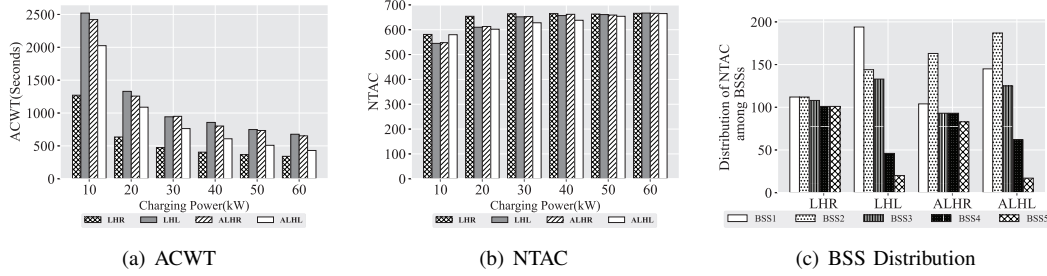


Fig. 11. FDI attack: Impact of charging power

insights into BSS service efficiency. In our analysis, we assume sufficient availability of CPUs to evaluate the operational dynamics at BSSs, and attackers manipulate local status data of BSS_2 in Fig. 7.

1) *Impact of AEV Density*: Fig. 10(a) demonstrates that the ACWT increases across all schemes as the density of AEVs rises, a trend consistent with the findings in Fig. 8(a). This pattern remains unchanged regardless of an attack, indicating a direct correlation between AEV density and ACWT. Since each BSS has a finite number of stock batteries, energy supplement demands are fully satisfied at lower AEV densities (e.g., 200 or 300 AEVs). Fig. 10(b) shows that the NTAC remains fairly consistent across all schemes. However, our proposed LHR strategy consistently outperforms others, especially at higher AEV densities, by achieving the lowest ACWT. This advantage is attributed to effective network load balancing, as depicted in Fig. 10(c), showing NTAC distribution across BSSs for 600 AEVs. It can be observed that the LHL, ALHR and ALHL schemes are in a quite skewed distribution. Especially, the local status of BSS_2 is attacked and a service hotspot is generated artificially, further increasing queuing time. By contrast, the service load across the network can be effectively balanced with the proposed LHR scheme.

2) *Impact of Charging Power*: Fig. 11 assesses the influence of charging power at BSSs, comparing various schemes under an attack scenario. Fig. 11(a) demonstrates that the improvement of charging power substantially reduces the ACWT. When an AEV arrives at a BSS without available batteries, it has to wait for depleted batteries completing charging. Thus, a higher charging power naturally shortens this duration, resulting in a reduced ACWT. Meanwhile, more AEV would be serviced with increased charging power, as shown in Fig. 11(b). Regarding a setup with 60 kW charging power and a default network scale of 300 AEVs, Fig. 11(c) illustrates the distribution of NTAC across BSSs. Although Fig. 11(c) exhibits a similar tendency with Fig. 10(c) overall,

the LHR presents more balanced distribution in Fig. 11(c). This is because the larger AEV network size (600 AEVs), increasing simultaneous battery swapping demands. Despite the LHR scheme can predicate the potential hotspot, it may also recommend AEVs with concurrent requirement to the same BSS, thus causing a more skewed distribution than a smaller scale of AEVs. But even so, it also exhibits superior performance than other related schemes significantly.

In the scenario of research on FDI Attack, for above all related algorithms, the timing analysis is presented in Table III. The time complexity of detecting the FDI attack is $O(nm^2)$ and the time complexity of removing the FDI attack is $O(m)$.

TABLE III
THE TIMING ANALYSIS RESULTS OF ALL RELATED ALGORITHMS

	LCD	LCR	LHR	LHL	ALHR	ALHL
Time Complexity	$O(ck)$	$O(cm^2)$	$O(kmn^2)$	$O(kmn)$	$O(km^2n^2)$	$O(km^2n)$

3) *Impact of Supply Chain Vulnerabilities*: For the FDI attack on supply chain vulnerabilities, the attacker manipulates the rated capacity of depleted batteries to a larger value, and the FDI attack is occurred at BSS_1 . The initial number of batteries at each BSS is set as 20.

With a variation of charging power, performance metrics based on supply chain vulnerabilities are presented in Fig. 12. As depicted in Fig. 12(a) and Fig. 12(b), for reservation scheme (i.e. LHR and ALHR), the impact of attack is relatively remarkable at 10 kW. With the increasement of charging power, the adverse effect of FDI attack can be alleviated. This is because the FDI attacker aiming to tamper the rated capacity of depleted batteries to a larger value, causing a longer service waiting time estimate value. However, the estimation of expected waiting time for swapping is implemented only when there are no available batteries at BSS. Benefiting from a high charging power at BSS, the depleted batteries can become available in a short time, and thus supply chain vulnerabilities will not have a remarkable influence on battery swapping. For

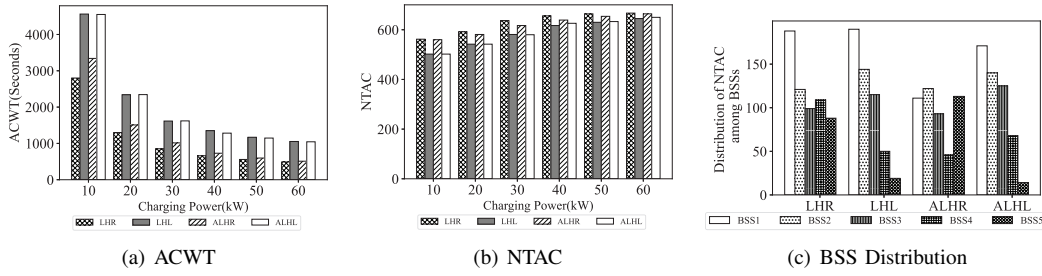


Fig. 12. FDI attack: Impact of Supply Chain Vulnerabilities with Variation of Charging Power

non-reservation schemes (i.e. LHL and ALHL), the effect of attack on ACWT is not obvious. This does not mean the FDI attack does not take effect. It is just because the ACWT of LHL scheme is awful enough. The effect of FDI attack can be clearly depicted in Fig. 12(c). For BSS_1 , a longer service waiting time estimate value due to supply chain vulnerabilities causes AEVs to prefer not to be serviced.

4) *The Case Study on a Single EV*: Different from above large-scale AEV service scenario simulation, we focus on a single AEV case study in this part. Other configurations are similar to the default setting. The simulation result presents the implementation process of proposed algorithm. Besides, the impact of FDI attack on energy supplement and parking decision-making strategy has been revealed in the case study. The result of selection strategy of parking lot and BSS is depicted in Table IV. Fig. 2 reflects the time sequence of energy supplement and parking decision-making strategy, and Table IV illustrates the specific implementation process of proposed algorithm. The impact of FDI attack has also been further investigated and compared with the no-attack situation.

TABLE IV
A SINGLE AEV CASE STUDY

Attack Status	SOC Status	LHR	ALHR
Without Attack	Above Threshold	$DP_2 \rightarrow CPU_4$	$DP_2 \rightarrow CPU_4$
	Below Threshold	$DP_2 \rightarrow BSS_5 \rightarrow CPU_4$	$DP_2 \rightarrow BSS_5 \rightarrow CPU_4$
Attack	Above Threshold	$DP_2 \rightarrow CPU_4$	$DP_2 \rightarrow CPU_4$
	Below Threshold	$DP_2 \rightarrow BSS_2 \rightarrow CPU_2$	$DP_2 \rightarrow BSS_5 \rightarrow CPU_4$

Specifically, when AEV is above the default SOC threshold without FDI attack, the parking requirement without energy supplement is primary concern. As depicted in Table IV, the LHR scheme will guide the AEV to drop off customer at DP_2 , and then park at CPU_4 . This is because CPU_4 is the nearest parking slot to DP_2 . By contrast, once AEV is below the default SOC threshold, the energy supplement is required. Thus, the optimal BSS_5 is recommended to AEV for battery swapping service due to its desirable position and available service status. After battery swapping operation, the AEV moves towards the nearest CPU_4 for parking. When AEV is above the default SOC threshold in FDI attack, the parking decision-making will not change. This is because the FDI attack is mainly carried out at BSS and affects the optimal BSS selection. Once below SOC threshold, the AEV should have moved to BSS_5 . However, the status information of BSS_2 is manipulated and a service advantage is generated artificially. Therefore, the AEV is misled to BSS_2 for energy supplement and then park at CPU_2 . The FDI attack does not directly affect AEV parking strategy but can further affect parking strategy by

changing the selection of optimal BSS. The LHR and ALHR schemes achieve the same decision-making in a no-attack environment. When facing FDI attack, the proposed ALHR scheme can detect the attack and guarantee the reliability of optimal BSS-selection, as depicted in TABLE IV.

D. Future Work

Considering the benefits of safety, authenticity, transparency and immutability, the blockchain technology can be regarded as a promising method to address supply chain vulnerabilities during battery swapping service involved whole production, sales, and service process. A blockchain-based battery tracking platform and smart contracts involved above supply chain entities are executed to decrease transaction costs, unnecessary testing and rework. During the detailed implementation process, a specific identifier is allocated to the battery. Users, inspection agents, or an authorized participant can thus obtain the battery information, such as the manufacturer and historical users. Through the designed blockchain-based platform, the whole process of battery circulation can be monitored.

In this paper, with the help of ICT, all EVs report their personal information (e.g., location information, historical service information, and trade data) to the GC in a centralized manner. GC can obtain all BSS operation data and status information. The privacy issue and the robustness are the main limitations of the paper. To overcome these limitations, cryptonyms or encryption methods can be applied in communication process. Besides, future work can investigate a distributed management framework based on Mobile Edge Computing (MEC) to improve system robustness.

VI. CONCLUSION

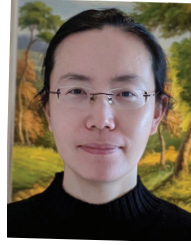
In this paper, we have developed a comprehensive LAVP management framework that integrates battery swapping service, so as to provide a coordinated decision-making optimization regarding energy supplementation convenience and parking favor for AEVs. To be specific, the proposed approach focuses on where to refill and park in the sense of a systematic approach. Once the battery level is below the SOC threshold, the GC preferentially recommends AEV to battery swapping service and park at CPU, aiming at timely response for subsequent pick-up demand from users. Considering the potential vulnerabilities of BSS system operator, we introduced a detection mechanism that monitors BSS status changes and validates data reliability through observation-value detection. In cases where a BSS is compromised, the proposed information recovery method accurately reflects the actual status of the BSS, allowing for reliable and optimal

decision-making in BSS selection. By comparing with other benchmarks, simulation results show the effectiveness of proposed scheme, in terms of alleviating charging congestion, timely demand response and mitigating the effect of attack.

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